Lexington Whalen

whalenlex[@]gmail.com · lxaw.github.io · linkedin.com/in/lxaw

Summary

Machine learning researcher specializing in efficient training and inference of large language models. Currently co-lead pretraining and inference optimization research for the Sarashina LLM series at SoftBank's SB Intuitions; previously co-led NVIDIA's flagship diffusion language models, building training and inference infrastructure adopted by 5+ internal research teams. Publications at ICML, CVPR, and ACL; NSF Graduate Research Fellow. Proven record of leading research teams in both the United States and Japan.

Experience

SoftBank – SB Intuitions — **Tokyo, Japan**
Pretraining & Inference Optimization Research Lead — *Apr. 2026 –*

- Co-lead pretraining and post-training optimization research for the Sarashina series, Japan's largest LLMs, at SoftBank's SB Intuitions R&D Division.
- Built SB Intuitions' first inference optimizations via speculative decoding, delivering significant throughput and latency improvements across the Sarashina model family.
- Serve as a key technical voice in Japanese-language cross-team discussions on pretraining, post-training, and inference optimization, bridging research and engineering.
- Collaborate with leading Japanese and international technology companies to evaluate GPU resources and co-develop long-term compute strategies for large-scale training.

NVIDIA — **Atlanta, GA & Santa Clara, CA**
Efficient Deep Learning Intern & Data Filtering Challenge Lead — *Dec. 2024 – Mar. 2026*

- Helped lead development of the Nemotron series of diffusion language models.
- Co-developed the EfficientDLM series of diffusion language models (base, instruct, and RL variants), outperforming the Qwen3 autoregressive series in both accuracy and speed (ICML 2026); work presented directly to CEO Jensen Huang.
- Built the training, evaluation, and inference infrastructure adopted by 5+ internal research teams for vision-language models, embeddings, and latent reasoning, with over 1 million internal downloads.
- Organized NVIDIA's international data filtering challenge across 30+ university and industry teams, coordinating GPU resource allocation with sponsors including Lambda Labs and Turing.

Georgia Institute of Technology — **Atlanta, GA**
Machine Learning Engineer — *Aug. 2024 – Dec. 2025*

- Led a team of 3 Ph.D. and Masters students to reduce diffusion model training time by up to 5x while maintaining generation quality; accepted to CVPR 2025.
- Algorithm lead for the United States' ARPA-H THOR healthcare initiative: reduced cancer detection sensor degradation by over 70%, extending sensor lifetime from a few hours to several days.
- Engineered resource-efficient ML architectures powered by energy from the human body, collaborating with teams from Stanford, Northwestern, Carnegie Mellon, and MIT.
- Presented technical progress to federal sponsors and healthcare stakeholders nationwide.

Selected Publications & Patents

- **L. Whalen**, Y. Ito, R. Sakamoto. "Teaching Diffusion to Speculate Left-to-Right." *Tech Report*, 2026. [SB Intuitions]
- Y. Fu, **L. Whalen**, A. Garg, C. Wu, M. Khadkevich, N. Oswald, E. Xie, D. Egert, S. Turuvekere Sreenivas, S. Diao, C. Yu, Y. Yu, W. Chen, S. Norouzi, S. Lan, L. Zhu, J. Wang, J. Jiang, M. Mardani, M. Maghoumi, S. Han, A. Jukić, N. Tajbakhsh, J. Kautz, P. Molchanov. "Nemotron-Labs-Diffusion: A Tri-Mode Language Model Unifying Autoregressive, Diffusion, and Self-Speculation Decoding." *Tech Report*, 2026. [NVIDIA]

- Y. Fu*, **L. Whalen***, Z. Ye, X. Dong, S. Diao, J. Liu, C. Wu, H. Zhang, E. Xie, S. Han, M. Khadkevich, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in Speed." *International Conference on Machine Learning (ICML)*, 2026. [NVIDIA]
- Y. Fu, **L. Whalen**, X. Dong, S. Diao, C. Wu, E. Xie, S. Han, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Training Framework for Converting Autoregressive Language Models into Faster Diffusion Language Models." *U.S. Patent, to be filed*, 2026. [NVIDIA]
- C. Shih, C. Li, C. Yu, H. Fang, S. Li, W. Hsin, **L. Whalen**, H. Suh, G. Eisenhauer, L. Liu, Y. (Celine) Lin. "NeRArch-Sim: A Unified Simulator for Benchmarking and DSE of Neural Rendering Accelerators." *International Symposium on Computer Architecture (ISCA)*, 2026. [Georgia Tech]
- **L. Whalen***, Z. Du*, H. You*, C. Li, S. Li, Y. (Celine) Lin. "Early-Bird Diffusion: Investigating and Leveraging Timestep-Aware Early-Bird Tickets in Diffusion Models for Efficient Training." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [Georgia Tech]
- Z. Ye, Z. Wang, K. Xia, J. Hong, L. Li, **L. A. Whalen**, C. Wan, H. You, C. Lin, S. Kundu. "LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering." *Association for Computational Linguistics (ACL)*, 2025. [Georgia Tech]
- A. Murphy, **L. Whalen**, S. Dubinsky, M. Gavin, J. F. Bailyn, J. Ginn. "On 'historical unity' of Russian and Ukrainian: A linguistic perspective on language conflict and change." *Linguistic Society of America (LSA)*, Apr. 2023. [University of South Carolina]

* Equal contribution

Invited Talks

- "拡散言語モデル：基本から応用まで", SoftBank – SB Intuitions — Jun. 2026
  Diffusion Language Models: From Fundamentals to Applications
- 入社式スピーチ, SoftBank — Apr. 2026
  Entrance Ceremony Speech

Education

Georgia Institute of Technology — **Atlanta, GA**
Graduate Researcher – Efficient Machine Learning Systems — *Aug. 2024 – Dec. 2025*

University of South Carolina, Honors College — **Columbia, SC**
Accelerated Master's in Computer Science — *Jan. 2021 – Aug. 2024*

Awards & Recognition

- SoftBank Group Representative, 2026 Entrance Ceremony — Mar. 2026
- NSF Graduate Research Fellowship ($150,000 value) — Apr. 2024
- Toyo University Exchange Student Representative — Jul. 2023
- University of South Carolina Outstanding Senior, Class of 2023 — Feb. 2023
- U.S. Department of State Critical Language Scholar of Japanese — Jan. 2023
- University of South Carolina Top Scholar ($100,000+ value) — Aug. 2020

Language Skills

- English – Native
- Japanese – Native Level (JLPT N1 Certified)
- Mandarin – Intermediate
- German – Beginner